Kinetic Theory of Gases

Syllabus Context: Molecular nature of matter, behaviour of ideal and real gases, gas laws, kinetic theory of an ideal gas, pressure and kinetic interpretation of temperature, law of equipartition of energy, specific heat capacities of gases and solids, and mean free path.

1. Learning Objectives & Overview

In 1661, Robert Boyle discovered the experimental law relating gas pressure and volume. Over the next two centuries, scientists including Boyle, Newton, Maxwell, and Boltzmann developed the **Kinetic Theory of Gases**, which explains macroscopic gas behaviour by modeling a gas as a large collection of rapidly moving atoms or molecules. Because intermolecular forces are short-range forces that are significant in solids and liquids, they can be largely neglected in gases except during momentary collisions.

Key Competencies for NEET UG

- Apply the ideal gas equation $PV = \mu RT = k_{\mathrm{B}}NT$ and Dalton's law of partial pressures to solve quantitative problems on gas state variables and mixtures.
- Relate the macroscopic pressure of an ideal gas to microscopic molecular parameters via $P = \frac{1}{3}nm\overline{v^2}$ and $PV = \frac{2}{3}E$.
- Compute root mean square speed $v_{\mathrm{rms}} = \sqrt{\frac{3k_{\mathrm{B}}T}{m}} = \sqrt{\frac{3RT}{M_0}}$ and average translational kinetic energy $\frac{3}{2}k_{\mathrm{B}}T$ per molecule.
- Determine degrees of freedom, internal energy U, molar specific heats (C_v and C_p), and adiabatic ratio $\gamma = \frac{C_p}{C_v}$ for monatomic, diatomic (rigid and vibrating), and polyatomic gases, as well as solids.
- Calculate the mean free path $l = \frac{1}{\sqrt{2}n\pi d^2}$ and collision time τ of gas molecules.

2. Molecular Nature of Matter

The physicist Richard Feynman remarked that if all scientific knowledge were destroyed in a catastrophe and only a single sentence could be passed on to future generations, it should be the **Atomic Hypothesis**: *"All things are made of atoms — little particles that move around in perpetual motion, attracting each other when they are a little distance apart, but repelling upon being squeezed into one another."*

Historical Perspective: Atomic Hypothesis in Ancient India and Greece

Long before modern chemistry, scholars in ancient India and Greece proposed that matter is not continuous:

- **Ancient India (Vaiseshika School):** Founded by **Kanada** (6th century B.C.), this school postulated that matter consists of eternal, indivisible, infinitesimal ultimate particles called *Paramanu*. It was argued that if matter could be subdivided endlessly, there would be no difference between a mustard seed and the Meru mountain. Four kinds of atoms were postulated—*Bhoomi* (Earth), *Ap* (water), *Tejas* (fire), and *Vayu* (air)—each having characteristic mass and attributes, while *Akasa* (space) was considered continuous and inert. Two atoms combined to form a diatomic molecule (*dvyanuka*) and three formed a triatomic molecule (*tryanuka*). In *Lalitavistara* (a 2nd century B.C. biography of the Buddha), the atomic size was estimated to be of the order of $10^{-10}\,\mathrm{m}$, remarkably close to modern values!
- **Ancient Greece: Democritus** (4th century B.C.) proposed that matter consists of physically indivisible particles (*atom* means "indivisible" in Greek) that differ in shape, size, and properties. Water atoms were imagined as smooth and round (allowing easy flow), earth atoms as rough and jagged (holding together to form hard solids), and fire atoms as thorny. These early ideas remained philosophical conjectures because they were not tested by quantitative experiments.

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Scientific Atomic and Molecular Theory

Modern scientific atomic theory was introduced by **John Dalton** to explain two fundamental chemical laws:

1. **Law of Definite Proportions:** Any given chemical compound has a fixed proportion by mass of its constituent elements.
2. **Law of Multiple Proportions:** When two elements combine to form more than one compound, for a fixed mass of one element, the masses of the other element are in the ratio of small integers.

Dalton proposed that the smallest constituents of an element are identical atoms, which combine in small whole-number ratios to form molecules of compounds. Shortly thereafter, two gas laws reinforced the molecular viewpoint:

- **Gay-Lussac's Law:** When gases combine chemically to yield another gas, their volumes bear a ratio of small integers.
- **Avogadro's Law (Hypothesis):** Equal volumes of all gases at the same temperature and pressure contain equal numbers of molecules.

Microscopic Scales in Solids, Liquids, and Gases

Modern instruments such as electron microscopes and scanning tunnelling microscopes allow direct imaging of atoms and molecules. Key microscopic length scales are:

- **Typical Atomic Size:** About $1\,\text{Å} = 10^{-10}\,\text{m}$.
- **Solids:** Atoms are tightly packed with interatomic spacing of a few ångströms (typically $\approx 2\,\text{Å}$) and oscillate about fixed mean positions.
- **Liquids:** Interatomic separation is also about $2\,\text{Å}$, so liquids are nearly as dense as solids, but atoms are not rigidly fixed and can move around, enabling liquids to flow.
- **Gases:** Intermolecular distances are in tens of ångströms (around $20\,\text{Å}$ to $40\,\text{Å}$ under ordinary conditions). The average distance a molecule travels between successive collisions—the **mean free path**—is of the order of thousands of ångströms.

In solids and liquids, closeness makes the interatomic force (which features long-range attraction at a few ångströms and strong short-range repulsion when squeezed closer) crucial. In gases, molecules are far apart and move freely except during brief collisions. Although a gas in a container appears static macroscopically, it is in **dynamic equilibrium**: molecules continuously collide and change individual velocities while macroscopic averages remain constant.

3. Behaviour of Gases and Gas Laws

At low pressures and high temperatures (well above liquefaction temperatures), intermolecular attractions and finite molecular volumes become negligible. Under these conditions, gases obey a universal relation connecting pressure P, volume V, and absolute temperature T:

$$PV = k_\text{B} N T$$

$$\frac{P_1 V_1}{N_1 T_1} = \frac{P_2 V_2}{N_2 T_2} = k_\text{B} = 1.38 \times 10^{-23}\,\text{J}\,\text{K}^{-1}$$

Here N is the number of molecules in the sample and k_B is the **Boltzmann constant**, which is identical for all gases. Notice that if P, V, and T are the same for two gases, then $N_1 = N_2$, which directly establishes **Avogadro's Hypothesis**.

At **Standard Temperature and Pressure (STP)**—defined here as $T = 273\,\text{K}$ ($0\,^\circ\text{C}$) and $P = 1\,\text{atm} = 1.013 \times 10^5\,\text{Pa}$—a volume of $22.4\,\text{litres}$ ($22.4 \times 10^{-3}\,\text{m}^3$) of any ideal gas contains one mole of molecules, equal to **Avogadro's number**:

$$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$$

Forms of the Ideal Gas Equation

Using the number of moles $\mu = \frac{M}{M_0} = \frac{N}{N_\text{A}}$ (where M is the total mass of the gas and M_0 is its molar mass) and defining the universal gas constant $R = N_\text{A} k_\text{B} = 8.314\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$, the ideal gas equation can be written in three equivalent forms:

1. **Molar Form:**

$$PV = \mu R T$$

2. **Number Density Form:**

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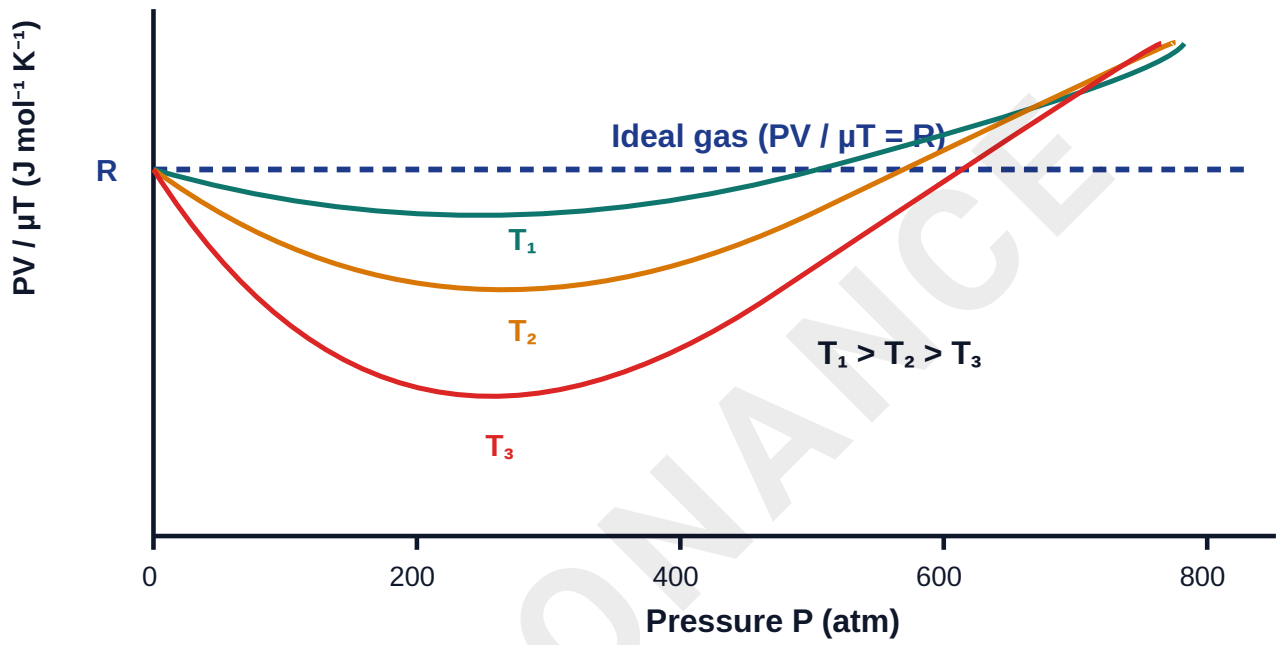
$$P = nk_{\mathrm{B}}T \quad \text{where } n = \frac{N}{V} \text{ is the number of molecules per unit volume.}$$

3. **Mass Density Form:**

$$P = \frac{\rho RT}{M_0} \quad \text{where } \rho = \frac{M}{V} \text{ is the mass density of the gas.}$$

Ideal versus Real Gases and Classical Gas Laws

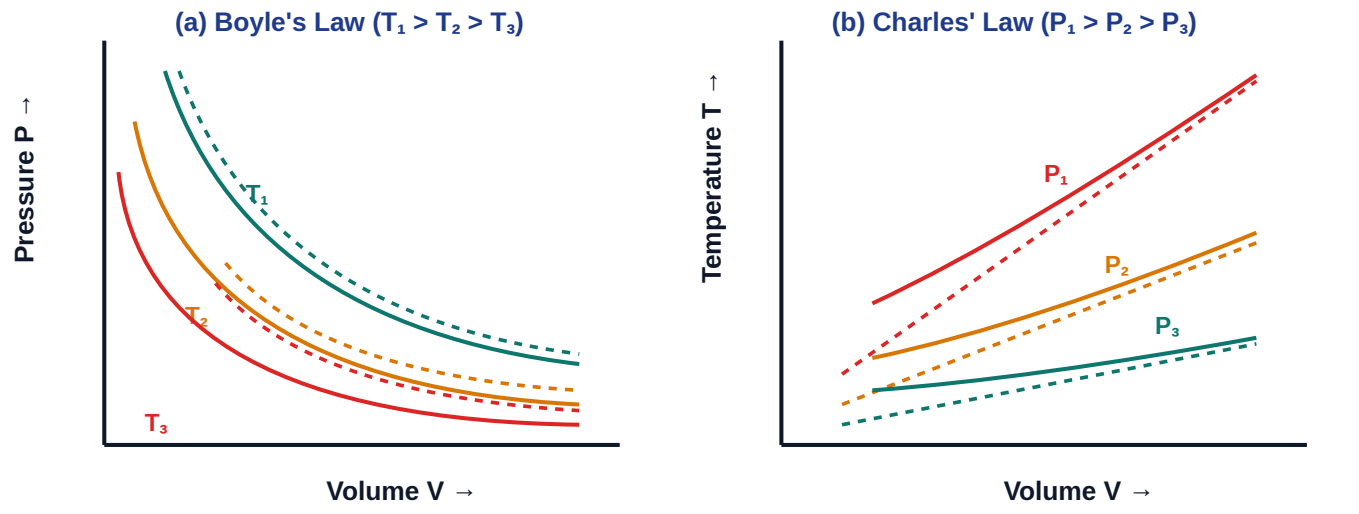
A gas that satisfies $PV = \mu RT$ at all pressures and temperatures is called an **ideal gas**. Real gases deviate from ideal behaviour at high pressures and low temperatures because intermolecular forces and finite molecular sizes can no longer be ignored. Real gases approach ideal gas behaviour at **low pressures and high temperatures**.



Tutorial Diagram 1: Departure of a real gas from ideal gas behaviour at three temperatures ($T_1 > T_2 > T_3$). All curves converge to R as $P \to 0$, and higher temperature curves stay closer to the horizontal ideal line.

- **Boyle's Law:** Keeping μ and T constant, $PV = \text{constant}$, or $P \propto \frac{1}{V}$. Experimental P–V curves for real gases (such as steam) agree closely with theoretical rectangular hyperbolas at high temperatures and low pressures.
- **Charles' Law:** Keeping μ and P constant, $\frac{V}{T} = \text{constant}$, or $V \propto T$. Experimental T–V plots for real gases (such as $\mathrm{CO_2}$) approach straight lines more closely at lower pressures.

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Tutorial Diagram 2: Comparison of experimental real-gas curves (solid lines) with theoretical ideal-gas predictions (dashed lines) for (a) Boyle's law ($T_1 > T_2 > T_3$) and (b) Charles' law ($P_1 > P_2 > P_3$).

Dalton's Law of Partial Pressures

Consider a mixture of non-interacting ideal gases containing μ_1 moles of gas 1, μ_2 moles of gas 2, and so on, enclosed in a vessel of volume V at absolute temperature T. The equation of state of the mixture is:

$$PV = (\mu_1 + \mu_2 + \cdots)RT$$

Dividing both sides by V:

$$P = \frac{\mu_1 RT}{V} + \frac{\mu_2 RT}{V} + \cdots = P_1 + P_2 + \cdots$$

Here $P_1 = \frac{\mu_1 RT}{V}$ is the **partial pressure** of gas 1—the pressure that gas 1 alone would exert if it occupied the entire volume V at temperature T in the absence of other gases. Thus, the total pressure of a mixture of non-reactive ideal gases equals the sum of the partial pressures of its constituent gases.

4. Kinetic Theory of an Ideal Gas

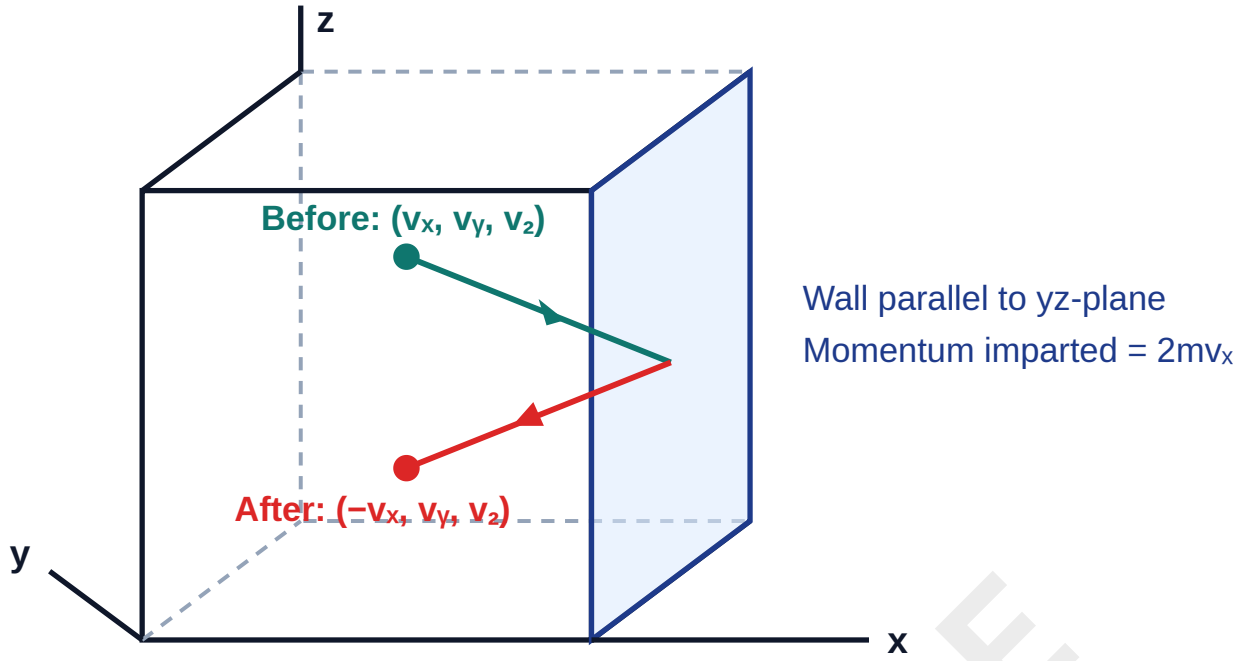
The kinetic theory of an ideal gas rests on the following microscopic postulates:

- A gas consists of a very large number of identical molecules (typically of the order of N_{A}) in incessant, random motion in all directions.
- At ordinary temperatures and pressures, the average distance between molecules is at least 10 times greater than the typical molecular diameter ($\approx 2\,\text{Å}$). Hence, intermolecular forces are negligible except during brief collisions, and molecules travel in straight lines at constant speed between collisions.
- All collisions—both between molecules and against the walls of the container—are **perfectly elastic**. Both total kinetic energy and total linear momentum are conserved, and the time spent during a collision is negligible compared to the time between collisions.
- The gas is **isotropic**: molecular velocities have no preferred direction in space.

4.1 Derivation of the Pressure of an Ideal Gas

Consider an ideal gas enclosed in a cubical vessel of side length l and wall area $A = l^2$, with coordinate axes chosen parallel to the cube edges. Suppose a molecule of mass m and velocity (v_x, v_y, v_z) strikes the planar wall parallel to the yz-plane.

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Tutorial Diagram 3: Elastic collision of a gas molecule with the container wall perpendicular to the x-axis. Only the normal velocity component v_x reverses sign.

Step 1 — Momentum transferred in one collision: Because the collision is elastic and the wall is stationary and massive, the parallel velocity components v_y and v_z remain unchanged while the normal component v_x reverses sign. The velocity after impact is $(-v_x, v_y, v_z)$. The change in linear momentum of the molecule along the x-axis is:

$$\Delta p_x = (-mv_x) - (mv_x) = -2mv_x$$

By conservation of momentum, the momentum imparted to the wall in a single collision is $+2mv_x$.

Step 2 — Number of collisions in time Δt: Within a small time interval Δt, any molecule with velocity component v_x located within a distance $v_x \Delta t$ from the wall (i.e., inside a volume $A v_x \Delta t$) can reach the wall, provided it is moving towards the wall. Because motion is random, on average exactly half of the molecules in this volume are moving towards the wall ($+x$) and half away from it ($-x$). If n is the number density of molecules, the number of molecules hitting the wall in time Δt is:

$$N_{\text{hit}} = \frac{1}{2} n A v_x \Delta t$$

Step 3 — Total momentum transfer and pressure: The total momentum Q transferred to the wall in time Δt by this group of molecules is:

$$Q = (2mv_x)\left(\frac{1}{2} n A v_x \Delta t\right) = n m v_x^2 A \Delta t$$

Since force is $F = \frac{Q}{\Delta t}$ and pressure is $P = \frac{F}{A} = \frac{Q}{A \Delta t}$, summing over the velocity distribution of all molecules gives:

$$P = nm\overline{v_x^2}$$

where $\overline{v_x^2}$ is the mean of the squared x-component of velocity. Because the gas is isotropic, there is no preferred direction in space:

$$\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2} = \frac{1}{3}\left(\overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2}\right) = \frac{1}{3}\overline{v^2}$$

where $\overline{v^2}$ (also written $\langle v^2 \rangle$) is the **mean square speed** of the molecules. Substituting this yields the fundamental pressure equation of kinetic theory:

$$P = \frac{1}{3} n m \overline{v^2} = \frac{1}{3} \rho \overline{v^2}$$

Important Remarks on the Derivation:

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- **Vessel shape does not matter:** Although derived for a cube, A and Δt cancel out. The same derivation applies to an infinitesimal planar area on a vessel of arbitrary shape, and by Pascal's law the pressure throughout a gas in equilibrium is uniform (ignoring gravity).
- **Intermolecular collisions do not alter the result:** Because the gas is in a steady state and collisions are random, if a molecule with velocity (v_x, v_y, v_z) is deflected before reaching the wall, another molecule acquires the same velocity (v_x, v_y, v_z) through collision, keeping the steady-state average $\overline{v_x^2}$ unchanged.

4.2 Kinetic Interpretation of Temperature and RMS Speed

Multiplying both sides of $P = \frac{1}{3}nm\overline{v^2}$ by volume V and noting that $N = nV$ is the total number of molecules in the sample:

$$PV = \frac{1}{3}Nm\overline{v^2} = \frac{2}{3}N\left(\frac{1}{2}m\overline{v^2}\right)$$

The total translational kinetic energy of all N molecules in the gas is $E = N\left(\frac{1}{2}m\overline{v^2}\right)$. Therefore:

$$PV = \frac{2}{3}E \quad \Longleftrightarrow \quad E = \frac{3}{2}PV = \frac{3}{2}\mu RT = \frac{3}{2}k_{\mathrm{B}}NT$$

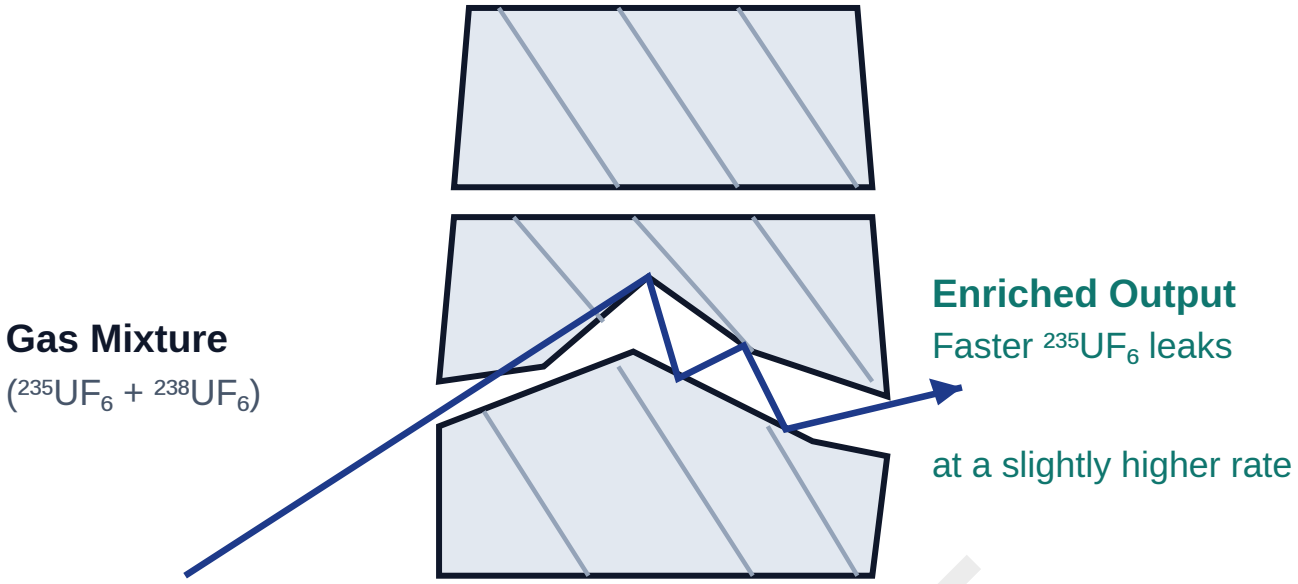
Dividing E by the total number of molecules N gives the **average translational kinetic energy per molecule**:

$$\frac{E}{N} = \frac{1}{2}m\overline{v^2} = \frac{3}{2}k_{\mathrm{B}}T$$

Key Conceptual Insights:

1. The average translational kinetic energy of a gas molecule depends **only on absolute temperature** T. It is completely independent of pressure, volume, molar mass, or atomicity (whether the gas is monatomic, diatomic, or polyatomic).
2. The square root of $\overline{v^2}$ is called the **root mean square (rms) speed**, $v_{\mathrm{rms}} = \sqrt{\overline{v^2}}$:
$$v_{\mathrm{rms}} = \sqrt{\frac{3k_{\mathrm{B}}T}{m}} = \sqrt{\frac{3RT}{M_0}} = \sqrt{\frac{3P}{\rho}}$$
3. At a given temperature, lighter molecules move faster: $v_{\mathrm{rms}} \propto \frac{1}{\sqrt{m}} \propto \frac{1}{\sqrt{M_0}}$. For nitrogen ($\mathrm{N_2}$, $M_0 = 28\,\mathrm{g\,mol^{-1}} = 28 \times 10^{-3}\,\mathrm{kg\,mol^{-1}}$, molecular mass $m = 4.65 \times 10^{-26}\,\mathrm{kg}$) at $T = 300\,\mathrm{K}$, $v_{\mathrm{rms}} = 516\,\mathrm{m\,s^{-1}}$, which is of the order of the speed of sound in air.
4. For a mixture of non-reactive gases in thermal equilibrium at temperature T, molecules of every constituent gas have the same average translational kinetic energy:
$$\frac{1}{2}m_1\overline{v_1^2} = \frac{1}{2}m_2\overline{v_2^2} = \frac{3}{2}k_{\mathrm{B}}T$$
Substituting this into $P = \frac{1}{3}\left(n_1m_1\overline{v_1^2} + n_2m_2\overline{v_2^2} + \cdots\right)$ yields $P = (n_1 + n_2 + \cdots)k_{\mathrm{B}}T$, confirming Dalton's law of partial pressures from kinetic theory.

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Tutorial Diagram 4: Diffusion of gas molecules through a narrow, thick porous wall. Because lighter molecules have a higher average speed ($\propto M_0^{-1/2}$), they wander through the long pores slightly faster, enabling isotope enrichment.

5. Degrees of Freedom & Law of Equipartition of Energy

The number of independent coordinates required to specify the configuration or position of a molecule, or equivalently the number of independent modes in which a molecule can store mechanical energy, is called its **degrees of freedom**.

5.1 Types of Degrees of Freedom

- **Translational Degrees of Freedom:** A molecule moving along a straight line has 1 translational degree of freedom; in a plane it has 2; and in three-dimensional space it has **3 translational degrees of freedom**, corresponding to kinetic energy terms:

$$\varepsilon_t = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2$$

  In thermal equilibrium at temperature T, isotropy gives:

$$\left\langle \frac{1}{2}mv_x^2 \right\rangle = \left\langle \frac{1}{2}mv_y^2 \right\rangle = \left\langle \frac{1}{2}mv_z^2 \right\rangle = \frac{1}{2}k_{\mathrm{B}}T$$

- **Rotational Degrees of Freedom:** A monatomic gas molecule (such as He, Ne, or Ar) has only 3 translational degrees of freedom. A diatomic molecule (such as $\mathrm{O_2}$ or $\mathrm{N_2}$), modeled as a rigid dumbbell rotator at moderate temperatures, can also rotate about **two independent axes** perpendicular to the internuclear axis. Rotation along the line joining the two atoms has a negligible moment of inertia and is not excited for quantum-mechanical reasons. Thus, a rigid diatomic molecule has **2 rotational degrees of freedom**:

$$\varepsilon_r = \frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2$$

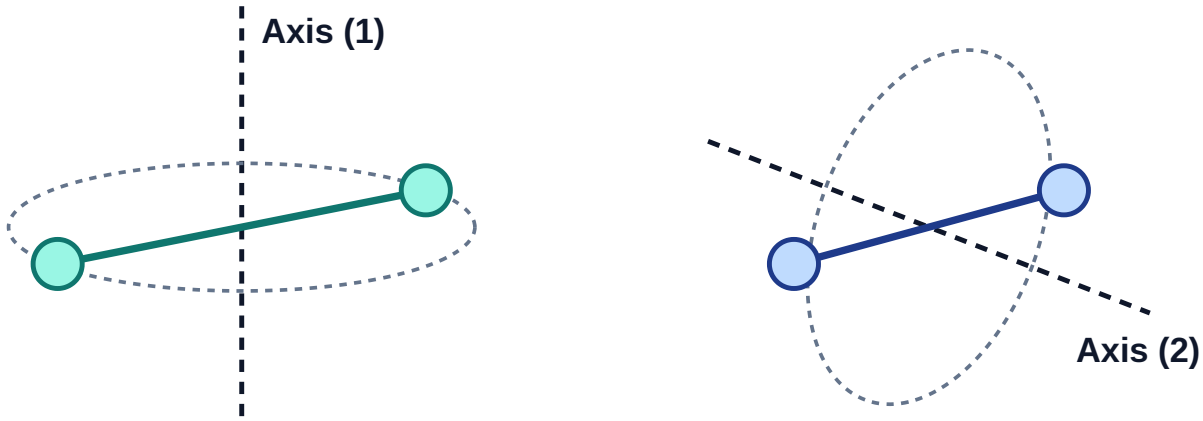
  where I_1, I_2 are moments of inertia and ω_1, ω_2 are angular speeds about the two perpendicular axes. A non-linear polyatomic molecule has **3 rotational degrees of freedom**.

- **Vibrational Modes:** Molecules that are not completely rigid (such as CO even at moderate temperatures, or $\mathrm{O_2}$ at high temperatures) oscillate along the interatomic bond like a one-dimensional harmonic oscillator. Each vibrational frequency (mode) contributes **two quadratic energy terms**—one kinetic and one potential:

$$\varepsilon_v = \frac{1}{2}m\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \frac{1}{2}ky^2$$

  where k is the force constant of the oscillator and y is the vibrational coordinate.

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Tutorial Diagram 5: The two independent axes of rotation (1) and (2) perpendicular to the internuclear axis of a diatomic molecule.

5.2 Statement of the Law of Equipartition of Energy

Law of Equipartition of Energy (Maxwell)

For any dynamical system in thermal equilibrium at absolute temperature T, the total energy is distributed equally among all possible quadratic energy modes of absorption, with each mode having an average energy equal to $\frac{1}{2}k_{\mathrm{B}}T$:

- Each **translational degree of freedom** contributes $\frac{1}{2}k_{\mathrm{B}}T$ per molecule.
- Each **rotational degree of freedom** contributes $\frac{1}{2}k_{\mathrm{B}}T$ per molecule.
- Each **vibrational mode (frequency)** contributes $2 \times \frac{1}{2}k_{\mathrm{B}}T = k_{\mathrm{B}}T$ per molecule (since it contains both a kinetic energy squared term and a potential energy squared term).

6. Specific Heat Capacities of Gases and Solids

Using the law of equipartition of energy, we can calculate the molar internal energy U of $1\ \mathrm{mole}$ (N_{A} molecules) of an ideal gas at temperature T, and from it obtain the molar specific heat at constant volume C_v, the molar specific heat at constant pressure C_p, and the adiabatic ratio γ:

$$C_v = \frac{\mathrm{d}U}{\mathrm{d}T}, \qquad C_p - C_v = R \quad (\text{Mayer's Relation, valid for any ideal gas}), \qquad \gamma = \frac{C_p}{C_v}$$

6.1 Monatomic, Diatomic, and Polyatomic Gases

- **Monatomic Gases (e.g., $\mathrm{He}, \mathrm{Ne}, \mathrm{Ar}$):** Only 3 translational degrees of freedom.

$$\text{Average energy per molecule} = \frac{3}{2}k_{\mathrm{B}}T \implies U = \frac{3}{2}k_{\mathrm{B}}TN_{\mathrm{A}} = \frac{3}{2}RT$$

$$C_v = \frac{3}{2}R, \qquad C_p = \frac{5}{2}R, \qquad \gamma = \frac{C_p}{C_v} = \frac{5}{3} \approx 1.67$$

- **Rigid Diatomic Gases (e.g., $\mathrm{O}_2, \mathrm{N}_2, \mathrm{H}_2$ at moderate temperatures):** 3 translational + 2 rotational = 5 degrees of freedom.

$$U = \frac{5}{2}k_{\mathrm{B}}TN_{\mathrm{A}} = \frac{5}{2}RT$$

$$C_v = \frac{5}{2}R, \qquad C_p = \frac{7}{2}R, \qquad \gamma = \frac{7}{5} = 1.40$$

- **Non-Rigid (Vibrating) Diatomic Gases (with 1 vibrational mode):** 3 translational + 2 rotational + 1 vibrational mode (which counts as 2 quadratic energy modes) = 7 energy modes.

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$$U = \left(\frac{5}{2}k_{\mathrm{B}}T + k_{\mathrm{B}}T\right)N_{\mathrm{A}} = \frac{7}{2}RT$$

$$C_v = \frac{7}{2}R, \qquad C_p = \frac{9}{2}R, \qquad \gamma = \frac{9}{7} \approx 1.29$$

- **Polyatomic Gases (Non-Linear with f vibrational modes):** A general non-linear polyatomic molecule has 3 translational degrees of freedom, 3 rotational degrees of freedom, and f vibrational modes (each contributing $k_{\mathrm{B}}T$):

$$U = \left(\frac{3}{2}k_{\mathrm{B}}T + \frac{3}{2}k_{\mathrm{B}}T + fk_{\mathrm{B}}T\right)N_{\mathrm{A}} = (3+f)RT$$

$$C_v = (3+f)R, \qquad C_p = (4+f)R, \qquad \gamma = \frac{4+f}{3+f}$$

If vibrational modes are ignored ($f = 0$), a non-linear triatomic/polyatomic gas has $C_v = 3R$, $C_p = 4R$, and $\gamma = \frac{4}{3} \approx 1.33$.

Nature of Gas (Ignoring Vibrations)	C_v **Formula**	C_v ($\mathrm{J\,mol^{-1}\,K^{-1}}$)	C_p **Formula**	C_p ($\mathrm{J\,mol^{-1}\,K^{-1}}$)	$C_p - C_v$ ($\mathrm{J\,mol^{-1}\,K^{-1}}$)	$\gamma = C_p/C_v$
Monatomic	$\frac{3}{2}R$	12.5	$\frac{5}{2}R$	20.8	8.31	1.67
Diatomic (Rigid)	$\frac{5}{2}R$	20.8	$\frac{7}{2}R$	29.1	8.31	1.40
Triatomic (Non-linear, $f = 0$)	$3R$	24.93	$4R$	33.24	8.31	1.33

Nature of Gas	**Gas**	**Measured** C_v ($\mathrm{J\,mol^{-1}\,K^{-1}}$)	**Measured** C_p ($\mathrm{J\,mol^{-1}\,K^{-1}}$)	**Measured** $C_p - C_v$ ($\mathrm{J\,mol^{-1}\,K^{-1}}$)	**Measured** γ
Monatomic	He	12.5	20.8	8.30	1.66
Monatomic	Ne	12.7	20.8	8.12	1.64
Monatomic	Ar	12.5	20.8	8.30	1.67
Diatomic	$\mathrm{H_2}$	20.4	28.8	8.45	1.41
Diatomic	$\mathrm{O_2}$	21.0	29.3	8.32	1.40
Diatomic	$\mathrm{N_2}$	20.8	29.1	8.32	1.40
Triatomic	$\mathrm{H_2O}$	27.0	35.4	8.35	1.31
Polyatomic	$\mathrm{CH_4}$	27.1	35.4	8.36	1.31

Notice that for polyatomic gases (and gases like $\mathrm{Cl_2}$ or $\mathrm{C_2H_6}$), experimental specific heats exceed the non-vibrating predictions because partial excitation of vibrational modes adds extra heat capacity.

6.2 Specific Heat Capacity of Solids (Dulong–Petit Law)

Consider a solid containing N atoms, each vibrating in three dimensions about its mean equilibrium position. Because a one-dimensional harmonic oscillation has two quadratic energy modes (kinetic and potential) with average energy $2 \times \frac{1}{2}k_{\mathrm{B}}T = k_{\mathrm{B}}T$, an atom vibrating in three independent dimensions has an average thermal energy of $3k_{\mathrm{B}}T$. For one mole of the solid ($N = N_{\mathrm{A}}$):

$$U = 3k_{\mathrm{B}}TN_{\mathrm{A}} = 3RT$$

Since the thermal expansion ΔV of a solid at constant atmospheric pressure is negligible, $\Delta Q = \Delta U + P\Delta V \approx \Delta U$. Therefore, the molar specific heat of a solid is:

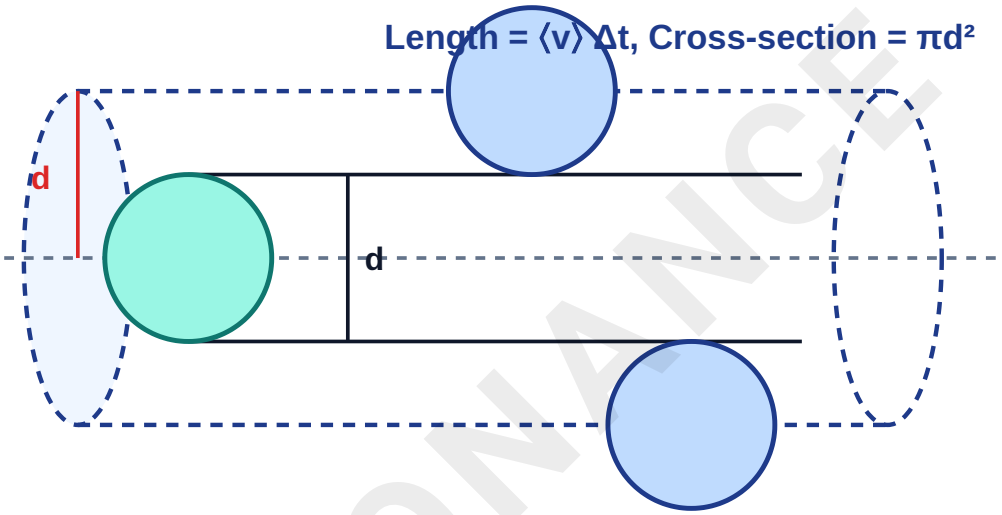
$$C = \frac{\Delta Q}{\Delta T} = \frac{\Delta U}{\Delta T} = 3R \approx 24.9\,\mathrm{J\,mol^{-1}\,K^{-1}}$$

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Solid Substance	Specific Heat ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	Molar Specific Heat ($\mathrm{J\,mol^{-1}\,K^{-1}}$)
Aluminium	900.0	24.4
Carbon	506.5	6.1 (Exception at room temperature)
Copper	386.4	24.5
Lead	127.7	26.5
Silver	236.1	25.5
Tungsten	134.4	24.9

7. Mean Free Path and Molecular Collisions

Although gas molecules travel at high speeds ($\sim 500\,\mathrm{m\,s^{-1}}$, comparable to the speed of sound), a leaking gas in a room takes considerable time to diffuse across the room because molecules have a finite diameter d and undergo frequent collisions that continually deflect their paths.



Tutorial Diagram 6: Collision cylinder of radius d and cross-sectional area πd^2 swept out by a spherical molecule of diameter d moving at average speed $\langle v \rangle$ in time Δt. Any molecule whose centre lies inside this cylinder collides with the moving molecule.

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Derivation of Collision Time and Mean Free Path

Model the molecules as hard spheres of diameter d. If a single molecule moves with average speed $\langle v \rangle$ while all other molecules are momentarily assumed stationary, it collides with any other molecule whose centre lies within a radial distance d of its path. In time Δt, it sweeps out a cylindrical volume of cross-sectional area πd^2 and length $\langle v \rangle \Delta t$:

$$V_{\text{swept}} = \pi d^2 \langle v \rangle \Delta t$$

If n is the number of molecules per unit volume, the number of collisions suffered in time Δt is $n\pi d^2 \langle v \rangle \Delta t$. Thus, the **collision frequency** (rate of collisions per unit time) is $f_c = n\pi d^2 \langle v \rangle$, and the average time between two successive collisions (**relaxation time or mean free time** τ) is:

$$\tau = \frac{1}{n\pi \langle v \rangle d^2} \quad \text{(stationary target approximation)}$$

Multiplying τ by the average speed $\langle v \rangle$ gives the approximate **mean free path** $l = \langle v \rangle \tau = \frac{1}{n\pi d^2}$.

Exact Treatment (Accounting for Relative Motion): In reality, all target molecules are also moving, so the collision rate is governed by the average relative speed $\langle v_{\text{rel}} \rangle = \sqrt{2}\langle v \rangle$ between colliding molecules. Replacing $\langle v \rangle$ by $\sqrt{2}\langle v \rangle$ yields the standard formula for the **mean free path**:

$$l = \frac{1}{\sqrt{2} n \pi d^2} = \frac{k_{\mathrm{B}} T}{\sqrt{2} \pi d^2 P}$$

Typical Values for Air at STP: With $n = 2.7 \times 10^{25}\,\mathrm{m^{-3}}$, $d = 2 \times 10^{-10}\,\mathrm{m}$, and $\langle v \rangle = 485\,\mathrm{m\,s^{-1}}$, we obtain $\tau \approx 6.1 \times 10^{-10}\,\mathrm{s}$ and $l \approx 2.9 \times 10^{-7}\,\mathrm{m} \approx 1500\,d$. Notice how the three fundamental microscopic length scales in a

gas at STP compare:

$$\text{Molecular diameter } d(\sim 2\,\text{Å}) \ll \text{Intermolecular distance } (\sim 30\text{–}40\,\text{Å}) \ll \text{Mean free path } l(\sim 2900\,\text{Å})$$

8. Comprehensive Worked Examples

Worked Example 1: Molecular Volume, Radius, and Intermolecular Distance in Water

The density of liquid water is $1000\,\text{kg}\,\text{m}^{-3}$, and the density of water vapour at $100\,^{\circ}\text{C}$ and $1\,\text{atm}$ pressure is $0.6\,\text{kg}\,\text{m}^{-3}$. Taking Avogadro's number $N_{\text{A}} = 6 \times 10^{23}\,\text{mol}^{-1}$ and molar mass of water $M_0 = 18\,\text{g}\,\text{mol}^{-1} = 0.018\,\text{kg}\,\text{mol}^{-1}$:

(a) Estimate the ratio (fraction) of the molecular volume to the total volume occupied by water vapour.

(b) Estimate the volume and radius of a single water molecule.

(c) Estimate the average interatomic (intermolecular) distance in water vapour under these conditions.

Solution:

(a) Fractional molecular volume in water vapour: For a fixed mass of water, volume is inversely proportional to density. The ratio of vapour volume to liquid volume is:

$$\frac{V_{\text{vapour}}}{V_{\text{liquid}}} = \frac{\rho_{\text{liquid}}}{\rho_{\text{vapour}}} = \frac{1000}{0.6} = \frac{1}{6 \times 10^{-4}} \approx 1.67 \times 10^{3}$$

In the liquid phase, molecules are closely packed so the fraction of molecular volume to total liquid volume is approximately 1. Since the vapour volume is larger by a factor of $\frac{1}{6\times10^{-4}}$, the fraction of molecular volume to total vapour volume is:

$$f = 6 \times 10^{-4}$$

(b) Volume and radius of a single water molecule: Mass of one water molecule is:

$$m = \frac{M_0}{N_{\text{A}}} = \frac{0.018\,\text{kg}}{6 \times 10^{23}} = 3 \times 10^{-26}\,\text{kg}$$

Approximating the density of a water molecule by the density of bulk liquid water ($1000\,\text{kg}\,\text{m}^{-3}$):

$$V_{\text{molecule}} = \frac{m}{\rho_{\text{liquid}}} = \frac{3 \times 10^{-26}\,\text{kg}}{1000\,\text{kg}\,\text{m}^{-3}} = 3 \times 10^{-29}\,\text{m}^3$$

Setting $V_{\text{molecule}} = \frac{4}{3}\pi r^3 = 3 \times 10^{-29}\,\text{m}^3$ gives the molecular radius:

$$r \approx 2 \times 10^{-10}\,\text{m} = 2\,\text{Å}$$

(c) Average intermolecular distance in water vapour: Since the volume available per molecule in the vapour phase increases by a factor of $1.67 \times 10^3 \approx 10^3$, the linear dimension (effective radius per molecule) increases by a factor of $(10^3)^{1/3} = 10$, from $2\,\text{Å}$ to $20\,\text{Å}$. Thus, the average distance between the centres of neighbouring water molecules in the vapour state is $2 \times 20\,\text{Å} = 40\,\text{Å}$.

BOSONANCE

Worked Example 2: Gas Mixture Partial Pressures and Mass Densities

A vessel contains a non-reactive mixture of neon (monatomic, atomic mass $M_1 = 20.2\,\text{u}$) and oxygen (diatomic, molecular mass $M_2 = 32.0\,\text{u}$). The ratio of their partial pressures is $\frac{P_1}{P_2} = \frac{3}{2}$. Find:
(i) the ratio of the number of molecules $\frac{N_1}{N_2}$, and
(ii) the ratio of mass densities $\frac{\rho_1}{\rho_2}$ of neon and oxygen in the vessel.

Solution:

(i) Ratio of number of molecules: Both gases occupy the same volume V at the same temperature T. Applying $P_1V = \mu_1 RT$ and $P_2V = \mu_2 RT$:

$$\frac{N_1}{N_2} = \frac{\mu_1 N_\text{A}}{\mu_2 N_\text{A}} = \frac{\mu_1}{\mu_2} = \frac{P_1}{P_2} = \frac{3}{2}$$

(ii) Ratio of mass densities: Since $\mu_1 = \frac{m_1}{M_1}$ and $\mu_2 = \frac{m_2}{M_2}$, and both gases occupy the same volume V:

$$\frac{\rho_1}{\rho_2} = \frac{m_1/V}{m_2/V} = \frac{m_1}{m_2} = \frac{\mu_1 M_1}{\mu_2 M_2} = \frac{3}{2} \times \frac{20.2}{32.0} = 0.947$$

Worked Example 3: Kinetic Energy and RMS Speeds in a Gas Mixture

A flask contains argon (atomic mass $39.9\,\text{u}$) and chlorine (molecular mass $70.9\,\text{u}$) in the ratio $2:1$ by mass at a temperature of $27\,^\circ\text{C}$. Obtain the ratio of:
(i) average translational kinetic energy per molecule, and
(ii) root mean square speed v_rms of the molecules of the two gases.

Solution:

(i) Ratio of average kinetic energy per molecule: The average translational kinetic energy per molecule of any ideal gas is $\frac{3}{2}k_\text{B}T$, which depends only on temperature T and is independent of the mass ratio or atomicity of the gases. Since both gases are in thermal equilibrium at $27\,^\circ\text{C}$ ($300\,\text{K}$), the ratio is:

$$\frac{\langle \varepsilon_t \rangle_\text{Ar}}{\langle \varepsilon_t \rangle_{\text{Cl}_2}} = 1:1$$

(ii) Ratio of root mean square speeds: Since $\frac{1}{2}mv_\text{rms}^2 = \frac{3}{2}k_\text{B}T$, we have $v_\text{rms}^2 \propto \frac{1}{M_0}$:

$$\frac{(v_\text{rms}^2)_\text{Ar}}{(v_\text{rms}^2)_{\text{Cl}_2}} = \frac{M_{\text{Cl}_2}}{M_\text{Ar}} = \frac{70.9}{39.9} = 1.77 \implies \frac{(v_\text{rms})_\text{Ar}}{(v_\text{rms})_{\text{Cl}_2}} = \sqrt{1.77} = 1.33$$

Note that the $2:1$ mass composition does not affect either ratio.

BOSONANCE

Worked Example 4: Isotope Enrichment via Molecular Speed Difference

Uranium has two isotopes of atomic masses $235\,\mathrm{u}$ and $238\,\mathrm{u}$. Both are present in uranium hexafluoride ($\mathrm{UF_6}$) gas, where fluorine has an atomic mass of $19\,\mathrm{u}$. Which molecule has the larger average speed at any given temperature, and what is the percentage difference in their speeds?

Solution:

First calculate the molecular masses of the two isotopic hexafluorides:

$$M(^{235}\mathrm{UF_6}) = 235 + 6 \times 19 = 235 + 114 = 349\,\mathrm{u}$$
$$M(^{238}\mathrm{UF_6}) = 238 + 6 \times 19 = 238 + 114 = 352\,\mathrm{u}$$

At a fixed temperature, average speed is inversely proportional to the square root of molecular mass ($v \propto M^{-1/2}$). Therefore, the lighter molecule $^{235}\mathrm{UF_6}$ ($349\,\mathrm{u}$) has the larger speed:

$$\frac{v_{349}}{v_{352}} = \left(\frac{352}{349}\right)^{1/2} = 1.0044$$

The fractional difference in speed is $\frac{\Delta v}{v} = 0.0044$, or 0.44%.

Worked Example 5: Elastic Collision with a Moving Piston and Adiabatic Heating

A molecule (or elastic ball) moving with speed u relative to a stationary cylinder wall approaches a massive piston (or heavy cricket bat) that is moving towards the molecule with speed V.
(a) Find the rebound speed of the molecule relative to the cylinder wall after an elastic collision.
(b) Use this result to explain microscopically why a gas heats up when compressed by pushing in a piston, and cools down when it expands by pushing a piston out.

Solution:

(a) Rebound speed: Relative to the approaching piston, the molecule approaches with speed $u_{\mathrm{rel}} = u + V$. Because the piston is massive and the collision is elastic, the molecule rebounds away from the piston with the same relative speed $u + V$ in the piston frame. Transforming back to the stationary cylinder frame (in which the piston is moving forward at speed V), the rebound speed of the molecule is:

$$v_{\mathrm{rebound}} = V + (u + V) = u + 2V$$

Thus, the molecule rebounds faster ($u + 2V > u$). (Similarly, in cricket, a heavy bat moving towards the ball transfers greater momentum and makes the ball rebound faster.)

(b) Microscopic explanation of compression heating and expansion cooling: When a gas is compressed by pushing a piston inward ($V > 0$ towards the gas), molecules colliding with the moving piston rebound with increased speed ($u + 2V$). This increases the mean square speed $\overline{v^2}$ of the gas molecules, which directly increases the absolute temperature T of the gas ($\frac{1}{2}m\overline{v^2} = \frac{3}{2}k_{\mathrm{B}}T$). Conversely, when a compressed gas pushes a piston outward during expansion, molecules hit a receding piston and rebound with a lower speed ($u - 2V$), reducing their average kinetic energy and causing the temperature of the gas to drop.

BOSONANCE

Worked Example 6: Heat Required for Isochoric Heating of Helium

A cylinder of fixed capacity $44.8\,\text{litres}$ contains helium gas at standard temperature ($273\,\text{K}$) and pressure ($1\,\text{atm}$). What amount of heat is needed to raise the temperature of the gas in the cylinder by $15.0\,^{\circ}\text{C}$? (Take $R = 8.31\,\text{J mol}^{-1}\,\text{K}^{-1}$.)

Solution:

At STP, $1\,\text{mol}$ of any ideal gas occupies the universal molar volume of $22.4\,\text{litres}$. Hence, a volume of $44.8\,\text{litres}$ at STP contains:

$$\mu = \frac{44.8\,\text{L}}{22.4\,\text{L mol}^{-1}} = 2\,\text{mol}$$

Helium is a monatomic gas, so its molar specific heat at constant volume is $C_v = \frac{3}{2}R = 1.5R$. Because the cylinder has fixed capacity (constant volume), the heat required for a temperature rise $\Delta T = 15.0\,^{\circ}\text{C} = 15.0\,\text{K}$ is:

$$Q = \mu C_v \Delta T = 2 \times (1.5R) \times 15.0 = 45R = 45 \times 8.31\,\text{J} = 374\,\text{J}$$

Worked Example 7: Mean Free Path of Water Vapour at $373\,\text{K}$

Estimate the mean free path for a water molecule in water vapour at $373\,\text{K}$ and $1\,\text{atm}$ pressure, given that at STP ($273\,\text{K}$, $1\,\text{atm}$) the number density of an ideal gas is $n_0 = 2.7 \times 10^{25}\,\text{m}^{-3}$ and the molecular diameter is $d = 2 \times 10^{-10}\,\text{m}$.

Solution:

At constant pressure ($1\,\text{atm}$), the number density $n = \frac{P}{k_B T}$ is inversely proportional to absolute temperature T. Therefore, at $T = 373\,\text{K}$:

$$n = 2.7 \times 10^{25} \times \frac{273}{373} \approx 2 \times 10^{25}\,\text{m}^{-3}$$

Substituting $n = 2 \times 10^{25}\,\text{m}^{-3}$ and $d = 2 \times 10^{-10}\,\text{m}$ into the mean free path formula:

$$l = \frac{1}{\sqrt{2} n \pi d^2} = \frac{1}{1.414 \times (2 \times 10^{25}\,\text{m}^{-3}) \times 3.14 \times (2 \times 10^{-10}\,\text{m})^2} \approx 4 \times 10^{-7}\,\text{m}$$

Note that $l = 4 \times 10^{-7}\,\text{m} = 4000\,\text{Å}$ is 100 times larger than the average intermolecular distance ($\sim 40\,\text{Å}$) in water vapour at $373\,\text{K}$.

BOSONANCE

Worked Example 8: Mass of Oxygen Withdrawn from a Pressurized Cylinder

An oxygen cylinder of volume $30\,\text{litres}$ has an initial gauge pressure of $15\,\text{atm}$ and a temperature of $27\,^\circ\text{C}$. After some oxygen is withdrawn from the cylinder, the gauge pressure drops to $11\,\text{atm}$ and its temperature drops to $17\,^\circ\text{C}$. Estimate the mass of oxygen taken out of the cylinder. (Take $1\,\text{atm} = 1.013 \times 10^5\,\text{Pa}$, $R = 8.31\,\text{J mol}^{-1}\,\text{K}^{-1}$, and molecular mass of $\text{O}_2 = 32\,\text{u}$).

Solution:

Step 1 — Convert gauge pressure to absolute pressure: Absolute pressure $P = P_{\text{gauge}} + P_{\text{atm}}$.

- Initial state: $P_1 = (15 + 1)\,\text{atm} = 16\,\text{atm} = 16 \times 1.013 \times 10^5\,\text{Pa} = 1.621 \times 10^6\,\text{Pa}$, $T_1 = 27 + 273 = 300\,\text{K}$, $V = 30\,\text{L} = 30 \times 10^{-3}\,\text{m}^3$.
- Final state: $P_2 = (11 + 1)\,\text{atm} = 12\,\text{atm} = 12 \times 1.013 \times 10^5\,\text{Pa} = 1.216 \times 10^6\,\text{Pa}$, $T_2 = 17 + 273 = 290\,\text{K}$.

Step 2 — Calculate initial and final moles:

$$\mu_1 = \frac{P_1 V}{R T_1} = \frac{1.621 \times 10^6 \times 30 \times 10^{-3}}{8.31 \times 300} = 19.51\,\text{mol}$$

$$\mu_2 = \frac{P_2 V}{R T_2} = \frac{1.216 \times 10^6 \times 30 \times 10^{-3}}{8.31 \times 290} = 15.14\,\text{mol}$$

Step 3 — Mass of oxygen withdrawn: With molar mass $M_0 = 32\,\text{g mol}^{-1} = 0.032\,\text{kg mol}^{-1}$:

$$\Delta m = (\mu_1 - \mu_2) M_0 = (19.51 - 15.14) \times 0.032\,\text{kg} = 4.37 \times 0.032\,\text{kg} \approx 0.140\,\text{kg} = 140\,\text{g}$$

Worked Example 9: Expansion of a Rising Air Bubble in a Lake

An air bubble of volume $V_1 = 1.0\,\text{cm}^3$ rises from the bottom of a lake $40\,\text{m}$ deep at a temperature of $12\,^\circ\text{C}$. To what volume V_2 does it grow when it reaches the surface, which is at a temperature of $35\,^\circ\text{C}$? (Take atmospheric pressure $P_0 = 1.013 \times 10^5\,\text{Pa}$, density of water $\rho_w = 1000\,\text{kg m}^{-3}$, and $g = 9.8\,\text{m s}^{-2}$).

Solution:

- At the bottom of the lake ($h = 40\,\text{m}$, $T_1 = 12 + 273 = 285\,\text{K}$):
  $P_1 = P_0 + \rho_w g h = 1.013 \times 10^5 + (1000 \times 9.8 \times 40) = 1.013 \times 10^5 + 3.92 \times 10^5 = 4.933 \times 10^5\,\text{Pa}$
- At the lake surface ($T_2 = 35 + 273 = 308\,\text{K}$):
  $$P_2 = P_0 = 1.013 \times 10^5\,\text{Pa}$$

Since the number of moles of air in the bubble remains constant, $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$:

$$V_2 = V_1 \left(\frac{P_1}{P_2}\right)\left(\frac{T_2}{T_1}\right) = 1.0\,\text{cm}^3 \times \left(\frac{4.933 \times 10^5}{1.013 \times 10^5}\right) \times \left(\frac{308}{285}\right) \approx 5.26\,\text{cm}^3$$

9. NEET UG Practice Questions with Worked Solutions

The following original practice questions are strictly aligned with the NEET UG physics examination format and syllabus.

BOSONANCE

Practice Question 1 (Single-Correct MCQ)

Estimate the fraction of molecular volume to the actual volume occupied by oxygen gas at STP ($T = 273\,\mathrm{K}$, $P = 1.013 \times 10^5\,\mathrm{Pa}$), taking the diameter of an oxygen molecule to be $3\,\text{Å}$.

(A) 3.8×10^{-4}

(B) 1.9×10^{-3}

(C) 7.6×10^{-2}

(D) 4.5×10^{-5}

Correct Answer: (A)

Worked Solution: The radius of one oxygen molecule is $r = \frac{d}{2} = 1.5\,\text{Å} = 1.5 \times 10^{-10}\,\mathrm{m}$. For $1\,\mathrm{mol}$ ($N_\mathrm{A} = 6.02 \times 10^{23}$ molecules) of oxygen at STP, the total volume occupied by the molecules themselves is:

$$V_\mathrm{mol} = N_\mathrm{A}\left(\frac{4}{3}\pi r^3\right) = 6.02 \times 10^{23} \times \frac{4}{3} \times 3.14 \times (1.5 \times 10^{-10}\,\mathrm{m})^3 \approx 8.51 \times 10^{-6}\,\mathrm{m}^3$$

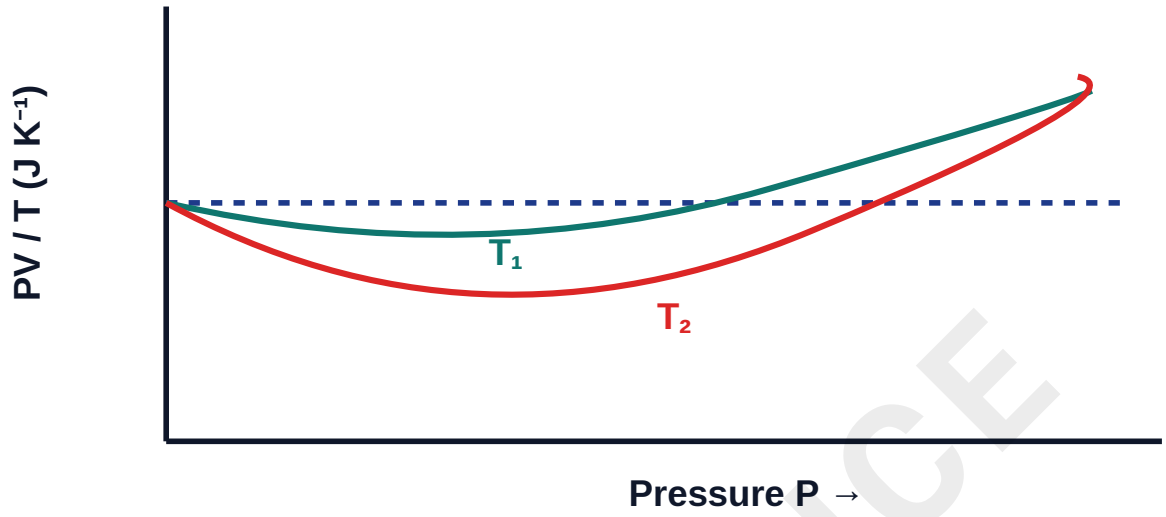
At STP, $1\,\mathrm{mol}$ of ideal gas occupies $V_\mathrm{actual} = 22.4\,\mathrm{litres} = 2.24 \times 10^{-2}\,\mathrm{m}^3$. Thus, the fraction is:

$$\frac{V_\mathrm{mol}}{V_\mathrm{actual}} = \frac{8.51 \times 10^{-6}\,\mathrm{m}^3}{2.24 \times 10^{-2}\,\mathrm{m}^3} \approx 3.8 \times 10^{-4}$$

BOSONANCE

Practice Question 2 (Graph-Based MCQ)

The graph below shows experimental plots of $\frac{PV}{T}$ versus pressure P for $1.00 \times 10^{-3}\,\text{kg}$ of oxygen gas ($M_0 = 32.0\,\text{u}$) at two temperatures T_1 and T_2, along with a horizontal dotted line. Take $R = 8.31\,\text{J mol}^{-1}\,\text{K}^{-1}$ and molecular mass of $\text{H}_2 = 2.02\,\text{u}$.



Which of the following combinations correctly states the temperature relation, the y-axis intercept of $\frac{PV}{T}$, and the mass of hydrogen gas (H_2) needed to produce the exact same y-axis intercept?

(A) $T_1 > T_2$; $\frac{PV}{T} = 0.26\,\text{J K}^{-1}$; mass of $\text{H}_2 = 6.3 \times 10^{-5}\,\text{kg}$

(B) $T_1 < T_2$; $\frac{PV}{T} = 0.26\,\text{J K}^{-1}$; mass of $\text{H}_2 = 1.00 \times 10^{-3}\,\text{kg}$

(C) $T_1 > T_2$; $\frac{PV}{T} = 8.31\,\text{J K}^{-1}$; mass of $\text{H}_2 = 6.3 \times 10^{-5}\,\text{kg}$

(D) $T_1 < T_2$; $\frac{PV}{T} = 0.52\,\text{J K}^{-1}$; mass of $\text{H}_2 = 3.2 \times 10^{-4}\,\text{kg}$

Correct Answer: (A)

Worked Solution:

1. The horizontal dotted line represents ideal gas behaviour ($\frac{PV}{T} = \mu R = \text{constant}$). Because a real gas is closer to an ideal gas at higher temperatures, the upper curve (closer to the dotted line) corresponds to the higher temperature: $T_1 > T_2$.
2. Where the curves meet on the y-axis ($P \to 0$), the gas behaves ideally:
$$\frac{PV}{T} = \mu R = \left(\frac{1.00 \times 10^{-3}\,\text{kg}}{32.0 \times 10^{-3}\,\text{kg mol}^{-1}}\right) \times 8.31\,\text{J mol}^{-1}\,\text{K}^{-1} \approx 0.26\,\text{J K}^{-1}$$
3. To obtain the same value of $\frac{PV}{T} = \mu R$, we need the same number of moles μ of H_2:
$$m_{\text{H}_2} = \mu M_{\text{H}_2} = \frac{1.00 \times 10^{-3}}{32.0} \times 2.02\,\text{kg} = 6.31 \times 10^{-5}\,\text{kg}$$

BOSONANCE

Practice Question 3 (Single-Correct MCQ)

Estimate the total number of air molecules (inclusive of oxygen, nitrogen, water vapour, and other constituents) in a room of capacity $25.0\,\mathrm{m}^3$ at a temperature of $27\,^\circ\mathrm{C}$ and $1\,\mathrm{atm}$ pressure ($1.013\times10^5\,\mathrm{Pa}$). (Take $k_\mathrm{B}=1.38\times10^{-23}\,\mathrm{J\,K^{-1}}$.)

(A) 6.1×10^{26}

(B) 2.5×10^{25}

(C) 6.02×10^{23}

(D) 9.2×10^{27}

Correct Answer: (A)

Worked Solution: Given $V=25.0\,\mathrm{m}^3$, $T=27+273=300\,\mathrm{K}$, and $P=1.013\times10^5\,\mathrm{Pa}$. Using $PV=k_\mathrm{B}NT$:

$$N=\frac{PV}{k_\mathrm{B}T}=\frac{1.013\times10^5\,\mathrm{Pa}\times25.0\,\mathrm{m}^3}{1.38\times10^{-23}\,\mathrm{J\,K^{-1}}\times300\,\mathrm{K}}\approx6.11\times10^{26}\ \text{molecules}$$

Practice Question 4 (Single-Correct MCQ)

What are the average thermal energies of a helium atom at (i) room temperature ($27\,^\circ\mathrm{C}$), (ii) the surface of the Sun ($6000\,\mathrm{K}$), and (iii) the core of a star ($10^7\,\mathrm{K}$)? ($k_\mathrm{B}=1.38\times10^{-23}\,\mathrm{J\,K^{-1}}$)

(A) $6.21\times10^{-21}\,\mathrm{J}$, $1.24\times10^{-19}\,\mathrm{J}$, and $2.07\times10^{-16}\,\mathrm{J}$

(B) $4.14\times10^{-21}\,\mathrm{J}$, $8.28\times10^{-20}\,\mathrm{J}$, and $1.38\times10^{-16}\,\mathrm{J}$

(C) $2.07\times10^{-21}\,\mathrm{J}$, $4.14\times10^{-20}\,\mathrm{J}$, and $6.90\times10^{-17}\,\mathrm{J}$

(D) $6.21\times10^{-20}\,\mathrm{J}$, $1.24\times10^{-18}\,\mathrm{J}$, and $2.07\times10^{-15}\,\mathrm{J}$

Correct Answer: (A)

Worked Solution: The average thermal (translational kinetic) energy of an atom at temperature T is $\langle E\rangle=\frac{3}{2}k_\mathrm{B}T$:

- (i) At $T=27\,^\circ\mathrm{C}=300\,\mathrm{K}$: $\langle E\rangle=1.5\times1.38\times10^{-23}\times300=6.21\times10^{-21}\,\mathrm{J}$.
- (ii) At $T=6000\,\mathrm{K}$: $\langle E\rangle=1.5\times1.38\times10^{-23}\times6000=1.242\times10^{-19}\,\mathrm{J}$.
- (iii) At $T=10^7\,\mathrm{K}$: $\langle E\rangle=1.5\times1.38\times10^{-23}\times10^7=2.07\times10^{-16}\,\mathrm{J}$.

BOSONANCE

Practice Question 5 (Multi-Statement Conceptual MCQ)

Three vessels of equal capacity contain gases at the same temperature and pressure. Vessel 1 contains neon (monatomic), Vessel 2 contains chlorine (diatomic), and Vessel 3 contains uranium hexafluoride (polyatomic). Consider the following statements:
Statement I: All three vessels contain the same number of respective molecules.
Statement II: The root mean square speed v_{rms} of molecules is the same in all three vessels.
Statement III: Neon molecules have the largest v_{rms} among the three vessels.
Which of the statements given above are correct?

(A) Statements I and III only

(B) Statements I and II only

(C) Statements II and III only

(D) Statements I, II, and III

Correct Answer: (A)

Worked Solution: By Avogadro's law ($N = \frac{PV}{k_{\text{B}}T}$), equal volumes of all ideal gases at the same T and P contain equal numbers of molecules regardless of atomicity, so Statement I is correct. However, $v_{\text{rms}} = \sqrt{\frac{3k_{\text{B}}T}{m}}$ depends inversely on the square root of molecular mass m. Since the molecular masses of Ne, Cl_2, and UF_6 are different, v_{rms} is not the same (Statement II is false). Because neon has the smallest molecular mass among the three, $(v_{\text{rms}})_{\text{Ne}}$ is the largest (Statement III is correct).

Practice Question 6 (Single-Correct MCQ)

At what temperature is the root mean square speed of an atom in an argon gas cylinder equal to the rms speed of a helium gas atom at $-20\,^{\circ}\text{C}$? (Atomic mass of $\text{Ar} = 39.9\,\text{u}$, atomic mass of $\text{He} = 4.0\,\text{u}$).

(A) $2.52 \times 10^3\,\text{K}$

(B) $253\,\text{K}$

(C) $1.26 \times 10^3\,\text{K}$

(D) $5.04 \times 10^3\,\text{K}$

Correct Answer: (A)

Worked Solution: Given $T_{\text{He}} = -20 + 273 = 253\,\text{K}$. Equating the root mean square speeds $v_{\text{rms}} = \sqrt{\frac{3RT}{M_0}}$ of argon and helium:

$$\frac{T_{\text{Ar}}}{M_{\text{Ar}}} = \frac{T_{\text{He}}}{M_{\text{He}}} \implies T_{\text{Ar}} = T_{\text{He}}\left(\frac{M_{\text{Ar}}}{M_{\text{He}}}\right) = 253\,\text{K} \times \frac{39.9}{4.0} = 2523.7\,\text{K} \approx 2.52 \times 10^3\,\text{K}$$

BOSONANCE

Practice Question 7 (Single-Correct MCQ)

A cylinder contains nitrogen gas ($\mathrm{N_2}$, molecular mass $28.0\,\mathrm{u}$) at a pressure of $2.0\,\mathrm{atm}$ ($2.026\times10^{5}\,\mathrm{Pa}$) and temperature $17\,^{\circ}\mathrm{C}$ ($290\,\mathrm{K}$). Taking the radius of a nitrogen molecule to be $r = 1.0\,\text{Å}$, estimate the mean free path l of a nitrogen molecule.

(A) $1.1\times10^{-7}\,\mathrm{m}$

(B) $4.5\times10^{-7}\,\mathrm{m}$

(C) $2.2\times10^{-6}\,\mathrm{m}$

(D) $5.5\times10^{-8}\,\mathrm{m}$

Correct Answer: (A)

Worked Solution: Diameter of the nitrogen molecule is $d = 2r = 2.0\,\text{Å} = 2.0\times10^{-10}\,\mathrm{m}$. First find the number density n:

$$n = \frac{P}{k_{\mathrm{B}}T} = \frac{2.026\times10^{5}\,\mathrm{Pa}}{1.38\times10^{-23}\,\mathrm{J\,K^{-1}}\times290\,\mathrm{K}} \approx 5.06\times10^{25}\,\mathrm{m^{-3}}$$

Now calculate the mean free path l:

$$l = \frac{1}{\sqrt{2}n\pi d^{2}} = \frac{1}{1.414\times(5.06\times10^{25}\,\mathrm{m^{-3}})\times3.14\times(2.0\times10^{-10}\,\mathrm{m})^{2}} \approx 1.11\times10^{-7}\,\mathrm{m}$$

(With $v_{\mathrm{rms}} = \sqrt{\frac{3RT}{M_0}} \approx 508\,\mathrm{m\,s^{-1}}$, the mean time between collisions is $\tau = \frac{l}{v} \approx 2.2\times10^{-10}\,\mathrm{s}$, corresponding to a collision frequency of $\sim 4.6\times10^{9}\,\mathrm{s^{-1}}$, whereas the duration of a single collision $\frac{d}{v} \approx 3.9\times10^{-13}\,\mathrm{s}$ is about 500 times smaller than the free-flight time τ.)

Practice Question 8 (Single-Correct MCQ)

One mole of a monatomic ideal gas ($\gamma_1 = \frac{5}{3}$) is mixed with one mole of a rigid diatomic ideal gas ($\gamma_2 = \frac{7}{5}$) at room temperature. What is the ratio of specific heats $\gamma_{\mathrm{mix}} = \frac{C_{p,\mathrm{mix}}}{C_{v,\mathrm{mix}}}$ for the mixture?

(A) 1.50

(B) 1.53

(C) 1.40

(D) 1.67

Correct Answer: (A)

Worked Solution: For $\mu_1 = 1\,\mathrm{mol}$ of monatomic gas, $C_{v1} = \frac{3}{2}R$. For $\mu_2 = 1\,\mathrm{mol}$ of rigid diatomic gas, $C_{v2} = \frac{5}{2}R$. Total internal energy of the $2\,\mathrm{mol}$ mixture is $U = \mu_1 C_{v1}T + \mu_2 C_{v2}T = \left(\frac{3}{2}R + \frac{5}{2}R\right)T = 4RT$. Thus, the molar specific heat at constant volume of the mixture is:

$$C_{v,\mathrm{mix}} = \frac{\mu_1 C_{v1} + \mu_2 C_{v2}}{\mu_1 + \mu_2} = \frac{4R}{2} = 2R$$

Using $C_{p,\mathrm{mix}} = C_{v,\mathrm{mix}} + R = 3R$, the adiabatic ratio of the mixture is:

$$\gamma_{\mathrm{mix}} = \frac{C_{p,\mathrm{mix}}}{C_{v,\mathrm{mix}}} = \frac{3R}{2R} = 1.50$$

BOSONANCE

Practice Question 9 (Single-Correct MCQ)

A non-linear polyatomic ideal gas molecule has 3 translational degrees of freedom, 3 rotational degrees of freedom, and 2 active vibrational modes at a given temperature. According to the law of equipartition of energy, what are the values of C_v, C_p, and γ for this gas?

(A) $C_v = 5R,\ C_p = 6R,\ \gamma = \frac{6}{5}$

(B) $C_v = 4R,\ C_p = 5R,\ \gamma = \frac{5}{4}$

(C) $C_v = \frac{7}{2}R,\ C_p = \frac{9}{2}R,\ \gamma = \frac{9}{7}$

(D) $C_v = 3R,\ C_p = 4R,\ \gamma = \frac{4}{3}$

Correct Answer: (A)

Worked Solution: For a polyatomic gas with 3 translational, 3 rotational degrees of freedom, and f vibrational modes, each vibrational mode contributes $2 \times \frac{1}{2}k_{\mathrm{B}}T = k_{\mathrm{B}}T$ per molecule. Therefore:

$$C_v = (3+f)R = (3+2)R = 5R$$
$$C_p = (4+f)R = (4+2)R = 6R$$
$$\gamma = \frac{C_p}{C_v} = \frac{4+f}{3+f} = \frac{6}{5} = 1.20$$

Practice Question 10 (Assertion–Reason MCQ)

Assertion (A): For a diatomic ideal gas that has 1 active vibrational mode in addition to translational and rotational motion, the molar specific heat at constant volume is $C_v = \frac{7}{2}R$.
Reason (R): According to the law of equipartition of energy, each vibrational mode contributes two quadratic energy terms (kinetic energy and potential energy), giving an average vibrational energy of $2 \times \frac{1}{2}k_{\mathrm{B}}T = k_{\mathrm{B}}T$ per molecule.

(A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Correct Answer: (A)

Worked Solution: A rigid diatomic molecule has 3 translational and 2 rotational degrees of freedom, contributing $\frac{5}{2}k_{\mathrm{B}}T$ per molecule. When 1 vibrational mode is also active, the vibrational energy expression $\varepsilon_v = \frac{1}{2}m\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \frac{1}{2}ky^2$ contains two quadratic terms, contributing $2 \times \frac{1}{2}k_{\mathrm{B}}T = k_{\mathrm{B}}T$ per molecule. Thus $U = \left(\frac{5}{2}+1\right)RT = \frac{7}{2}RT$ and $C_v = \frac{7}{2}R$. Both A and R are true, and R directly explains A.

BOSONANCE

Practice Question 11 (Matching MCQ)

Match each physical quantity or system in **List I** with its correct expression in **List II** for an ideal gas or solid in thermal equilibrium at absolute temperature T:

List I (System / Quantity)	List II (Expression)
(P) Molar specific heat of a solid at room temperature	(1) $\frac{2}{3}E$
(Q) Molar specific heat C_p of a rigid diatomic gas	(2) $3R$
(R) Product of pressure and volume (PV) in terms of translational kinetic energy E	(3) $\frac{7}{2}R$
(S) Mean free path l of gas molecules of diameter d and number density n	(4) $\frac{1}{\sqrt{2}n\pi d^2}$

(A) P–2, Q–3, R–1, S–4

(B) P–3, Q–2, R–1, S–4

(C) P–2, Q–3, R–4, S–1

(D) P–1, Q–3, R–2, S–4

Correct Answer: (A)

Worked Solution:

- (P) For 1 mole of a solid with atoms vibrating in 3D, $U = 3RT \implies C = 3R$ → (2).
- (Q) For a rigid diatomic gas, $C_v = \frac{5}{2}R \implies C_p = C_v + R = \frac{7}{2}R$ → (3).
- (R) From kinetic theory, $PV = \frac{2}{3}E$ → (1).
- (S) Mean free path $l = \frac{1}{\sqrt{2}n\pi d^2}$ → (4).

Practice Question 12 (Single-Correct MCQ)

Four molecules of a gas have speeds of $200\,\mathrm{m\,s^{-1}}$, $400\,\mathrm{m\,s^{-1}}$, $600\,\mathrm{m\,s^{-1}}$, and $800\,\mathrm{m\,s^{-1}}$. What is the difference between their root mean square speed (v_{rms}) and their average speed ($\langle v \rangle$)?

(A) $48\,\mathrm{m\,s^{-1}}$

(B) $0\,\mathrm{m\,s^{-1}}$

(C) $100\,\mathrm{m\,s^{-1}}$

(D) $24\,\mathrm{m\,s^{-1}}$

Correct Answer: (A)

Worked Solution: This question illustrates the important principle that $\langle v^2 \rangle \neq (\langle v \rangle)^2$. First compute the average speed:

$$\langle v \rangle = \frac{200 + 400 + 600 + 800}{4} = 500\,\mathrm{m\,s^{-1}}$$

Next compute the mean square speed and root mean square speed:

$$\langle v^2 \rangle = \frac{200^2 + 400^2 + 600^2 + 800^2}{4} = \frac{(4 + 16 + 36 + 64) \times 10^4}{4} = 30 \times 10^4\,\mathrm{m^2\,s^{-2}}$$

$$v_{\mathrm{rms}} = \sqrt{30 \times 10^4} = 100\sqrt{30} \approx 548\,\mathrm{m\,s^{-1}}$$

Therefore, $v_{\mathrm{rms}} - \langle v \rangle = 548 - 500 = 48\,\mathrm{m\,s^{-1}}$.

BOSONANCE

10. Common Pitfalls & Points to Ponder

1. Confusing Translational Kinetic Energy E with Total Internal Energy U

The relation $PV = \frac{2}{3}E$ always refers to the **translational kinetic energy** $E = \frac{3}{2}\mu RT$ of the gas molecules, regardless of whether the gas is monatomic, diatomic, or polyatomic. By contrast, total internal energy $U = \frac{f_{\text{total}}}{2}\mu RT$ includes rotational and vibrational energies as well. Only for a monatomic ideal gas is $E = U$.

2. Counting a Vibrational Mode as 1 Degree of Freedom Instead of 2 Energy Modes

Each translational and rotational degree of freedom contributes one squared term ($\frac{1}{2}k_{\text{B}}T$), whereas each **vibrational mode** contributes **two squared terms**—one for vibrational kinetic energy $\frac{1}{2}m\left(\frac{\text{d}y}{\text{d}t}\right)^2$ and one for vibrational potential energy $\frac{1}{2}ky^2$—giving $2 \times \frac{1}{2}k_{\text{B}}T = k_{\text{B}}T$ per molecule (or RT per mole).

3. Confusing Intermolecular Distance with Mean Free Path

Do not exaggerate the intermolecular distance in a gas at STP: it is only about 10–20 times the interatomic spacing in solids and liquids (~ 20–$40\ \text{Å}$ vs. $2\ \text{Å}$). What is dramatically larger in a gas is the **mean free path** l, which is about 100 times the intermolecular distance and 1000–1500 times the molecular diameter ($\sim 3000\ \text{Å}$).

4. Forgetting Gauge Pressure vs. Absolute Pressure and Celsius vs. Kelvin

In all ideal gas law and kinetic theory equations ($PV = \mu RT$, $v_{\text{rms}} = \sqrt{\frac{3RT}{M_0}}$), P must be the **absolute pressure** ($P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}}$) and T must be the **absolute temperature in kelvin** ($T(\text{K}) = t(^\circ\text{C}) + 273$). Also convert molar mass M_0 from g mol^{-1} to kg mol^{-1} when using SI units.

Additional Conceptual Points to Ponder

- **Pressure exists throughout the fluid:** Gas pressure is not exerted only on the walls of the container. Any internal layer of gas is in mechanical equilibrium because equal pressure acts on both sides of the layer.
- **Why air molecules do not settle on the floor under gravity:** Because of their high rms speeds ($\sim 500\ \text{m s}^{-1}$) and incessant collisions, gravitational potential energy mgh for ordinary room heights is negligible compared to average molecular kinetic energy $\frac{1}{2}m\overline{v^2} = \frac{3}{2}k_{\text{B}}T$, producing only a tiny density increase near the floor.
- **Mean of square vs. square of mean:** $\langle v^2 \rangle$ is always greater than $(\langle v \rangle)^2$ whenever there is a spread in molecular speeds.

BOSONANCE

11. Concise Formula & Revision Sheet

Concept / Quantity	Key Formula	Important Notes / Constants
Ideal Gas Equation	$PV = \mu RT = k_{\mathrm{B}}NT$ or $P = nk_{\mathrm{B}}T = \frac{\rho RT}{M_0}$	$R = 8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}$, $k_{\mathrm{B}} = \frac{R}{N_{\mathrm{A}}} = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
Dalton's Law of Partial Pressures	$P = P_1 + P_2 + \cdots = (\mu_1 + \mu_2 + \cdots)\frac{RT}{V}$	Valid for non-reactive ideal gas mixtures
Kinetic Pressure of Ideal Gas	$P = \frac{1}{3}nm\overline{v^2} = \frac{1}{3}\rho\overline{v^2}$ and $PV = \frac{2}{3}E$	$E = N\left(\frac{1}{2}m\overline{v^2}\right) = \frac{3}{2}k_{\mathrm{B}}NT$ is total translational KE
Average Translational KE per Molecule	$\frac{1}{2}m\overline{v^2} = \frac{3}{2}k_{\mathrm{B}}T$	Depends only on T; independent of gas nature
Root Mean Square (rms) Speed	$v_{\mathrm{rms}} = \sqrt{\overline{v^2}} = \sqrt{\frac{3k_{\mathrm{B}}T}{m}} = \sqrt{\frac{3RT}{M_0}} = \sqrt{\frac{3P}{\rho}}$	At fixed T, $v_{\mathrm{rms}} \propto \frac{1}{\sqrt{M_0}}$
Equipartition of Energy	$\frac{1}{2}k_{\mathrm{B}}T$ per translational or rotational degree of freedom; $k_{\mathrm{B}}T$ per vibrational mode	Each quadratic term in molecule's energy receives $\frac{1}{2}k_{\mathrm{B}}T$
Monatomic Gas Specific Heats	$U = \frac{3}{2}RT, \quad C_v = \frac{3}{2}R, \quad C_p = \frac{5}{2}R, \quad \gamma = \frac{5}{3} \approx 1.67$	3 translational degrees of freedom
Rigid Diatomic Gas Specific Heats	$U = \frac{5}{2}RT, \quad C_v = \frac{5}{2}R, \quad C_p = \frac{7}{2}R, \quad \gamma = \frac{7}{5} = 1.40$	3 translational + 2 rotational degrees of freedom
Vibrating Diatomic Gas (1 Mode)	$U = \frac{7}{2}RT, \quad C_v = \frac{7}{2}R, \quad C_p = \frac{9}{2}R, \quad \gamma = \frac{9}{7} \approx 1.29$	Includes 1 vibrational mode ($= 2 \times \frac{1}{2}k_{\mathrm{B}}T$)
Polyatomic Gas (f Vibrational Modes)	$C_v = (3+f)R, \quad C_p = (4+f)R, \quad \gamma = \frac{4+f}{3+f}$	3 translational + 3 rotational + f vibrational modes
Molar Specific Heat of Solids	$U = 3RT, \quad C = 3R \approx 24.9\,\mathrm{J\,mol^{-1}\,K^{-1}}$	Atoms vibrate in 3D; carbon is an exception at room temp.
Mean Free Path & Collision Time	$l = \frac{1}{\sqrt{2}n\pi d^2} = \frac{k_{\mathrm{B}}T}{\sqrt{2}\pi d^2 P}, \quad \tau = \frac{l}{\langle v \rangle}$	d = molecular diameter; $n = \frac{N}{V}$ = number density

BOSONANCE